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Optimal Liquidation Strategies in Illiquid Markets

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Abstract

In this paper, we consider block trading strategies and characterize the times when a block trade is a popular choice. We also study the economic relevance of optimal liquidation strategies by calibrating a recent and realistic microstructure model with data from the Paris Stock Exchange. We distinguish between two cases: one in which the parameters are constant throughout the day and one in which they vary over time. We present and solve an optimization problem incorporating this realistic microstructure model. Our model endogenizes the trading periods required before a position is liquidated. A comparative static exercise demonstrates the realism of our model. We also examine the model for bearish and bullish beliefs, demonstrating that volatility plays a role in determining the speed of trade execution.

Keywords: Optimal trading strategy, liquidity risk, price impact, high frequency data, microstructure.

JEL classification: G12.

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1 Introduction

We determine the optimal trading strategy for an investor who wishes to trade a large position in a given asset, in the event that the trade affects the liquidity of the asset. Such a strategy proves to be useful for financial institutions or mutual funds, which may buy or sell quantities that represent a significant fraction of a firm's market capitalization. An investor may first consider trading a block in the upstairs market. Such a trade would be convenient, as it requires only one counterparty. On the other hand, there is a search cost associated with finding the counterparty. Moreover, there is the constraint that the negotiated price lies within a range given by the volume-weighted average price of the book. Alternatively, the investor may wish to trade through an electronic system. In this case, the investor is confronted with a trade-off between the discount that a large trade may have and the risk taken by a given execution strategy. If she decides to liquidate the whole position immediately, she will know the liquidation value, but she may face a large discount in the selling price. On the other hand, if she decides to liquidate in a smooth way by selling small quantities, she might have a small discount but face the risk that most of the trades are executed after prices have moved. This trade-off implies a liquidation strategy involving the trade size, the trade price, and the time it takes to liquidate the entire block.

This paper contributes to the literature by documenting the finding that the decision to make block trades depends on market conditions. In liquid markets, search costs are presumably low and block trading may be an efficient strategy. However, in illiquid markets (and subsequent to a crash, in particular) it may be difficult to find a counterparty. In such a situation, it may be optimal to channel the order through an electronic trading system that allows for parceled trades. There is a large body of literature that considers the impact of block trades on prices. See Chan and Lakonishok (1995, 1997), Chiyachantana et al. (2006a and b), Kavajecz and Keim (2005), as well as Keim and Madhavan (1996, 1997). However, the majority of this literature considers the impact of a block trade on prices, rather than the decision of how to trade a large order. Keim and Madhavan (1995) represents one exception.

Analysis of optimal trading through an electronic system requires a realistic microstructure model as well as an algorithm to trade. We consider an extension of Sadka (2006) as our microstructure model. This is a rich model that incorporates many features exhibited in the microstructure literature. Indeed, it crystallizes the earlier contribu-

tions of Glosten and Harris (1988), Brennan and Subrahmanyam (1996), Madhavan, Richardson, and Roomans (1997), as well as Huang and Stoll (1997). Our extension allows for an asymmetric response to purchases or sales. The incremental impact of a sale will differ from that of a purchase.¹

We build upon Almgren and Chriss (2001) in forming the optimal trading strategy. Their model incorporates time-varying, permanent and transitory price impacts of the order flow and of the direction of the order flow. The investor maximizes the expected total income of liquidation penalized by its variance, where the penalization factor represents the investor's degree of risk aversion. The original problem is quadratic. We are able to express it as a linear optimization problem and solve it numerically. The strategy also endogenizes the number of periods over which the position should be sold. We study the characteristics of the solution and establish that the optimal number of trading periods decreases with the permanent fixed price impact and the standard deviation of the process associated with the arrival of public information. It increases with the variable unitary cost of trade and the fixed cost per order. We study how the solution to the optimization problem responds to patterns in the permanent impact of the order flow on price. We establish that the investor liquidates larger quantities when the impact of the order flow on price is small. We also estimate the parameters of the price process when they are assumed to be constant throughout the day, and when they change hourly. Time-changing parameters allow us to deal with modified trading patterns during the close of the market, in particular.

Other research has also considered algorithms for optimal trading. For instance, Bertsimas and Lo (1998) use stochastic dynamic programming techniques, which minimize the expected cost of purchasing a large block of shares. Contrary to Almgren and Chriss (2001), they do not incorporate the volatility of the total cost of trading into the optimization problem.

Dubil (2002) and Hisata and Yamai (2000) use a setting similar to Almgren and Chriss (2001) and obtain the optimal total time of liquidation assuming a constant liquidation speed. Huberman and Stanzl (2000), assuming slightly different price dynamics than Almgren and Chriss (2001), find a closed-form solution to the optimal static selling strategy when the price impact of the trade size is constant, and a recursive solution

¹Explanations for such a differentiated pattern may be found in Keim and Madhavan (1996). See also Chiyachantana *et al.* (2006a and b).

when the price impact changes over time.

Mönch (2004) incorporates two microstructure features in the price function: the U-shape of intraday market liquidity and the resiliency of the order book. Then, assuming that price dynamics only incorporate permanent impacts, he determines the optimal liquidation strategy numerically. Contrary to other papers, he does not force the time between trades to be constant.

Our paper differs from Almgren and Chriss (2001), Dubil (2002), and Hisata and Yamai (2000) in various respects. First, it does not restrict the price impact of the trade direction and the trade size to be constant over the time. This difference is achieved by allowing the parameters representing these impacts to be time-varying. Second, using tick-by-tick data from the Paris Stock Exchange, we are able to estimate the parameters that characterize the price process. It is worth emphasizing that, in this context, only Mönch (2004) has estimated the parameters of his proposed model. He assumes two particular functional forms for the price and traded size functions. Then, using tick-by-tick stock data from the Helsinki Stock Exchange and the same specification for all the stocks, he estimates the parameters using the least squares method. We use a more advanced model stemming from the microstructure literature.

The rest of the paper is organized as follows. In Section 2, we discuss the decision to trade blocks versus channeling an order through an electronic system. In Section 3, we present the optimization problem and its solution. We also introduce the microstructure model describing the dynamics of price evolution and we present the estimation methodology. In Section 4, we estimate the model for various companies. We consider two cases: when the parameters describing the price dynamics are constant throughout the day and when they change hourly. For both cases, we then determine the optimal liquidation strategy and study the sensitivity of the solution to changes in the levels of the parameters. We also study how the optimal liquidation strategy behaves when the value of the parameters evolves according to several pre-established patterns. This strategy allows us to show that bullish or bearish sentiments may lead to more aggressive trading patterns. Section 5 provides concluding comments.

2 Trading in Blocks or Not?

When evaluating the decision to trade, a trader faces the question of whether to execute by contracting potential counterparts in the so-called upstairs market or whether the trade should be routed directly through an electronic trading system. This issue was also considered by Keim and Madhavan (1995).

A formal model of this decision requires specification of the costs and benefits of one market with respect to the other. In principle, the electronic market is easier to describe than the upstairs market. In the first case, many public trades are available in order to estimate the key parameters describing this market. In the second case, it appears more difficult to describe the trading process and infer the price process from scarce data. Recent theoretical models of search, such as Duffie, Gârleanu, and Pedersen (2005), require knowledge of empirical calibration that is not available for the French stock market. As a consequence, we focus on a statistical description of upstairs trades in order to gain more insights into the determinants of upstairs market trading. A trader may decide which trading strategy to use based on this description.

Through a reform made in 1994, the Paris Stock Exchange introduced the possibility of block trades. Only the largest and most liquid companies are eligible to perform block trades. In addition, there is a lower bound on what constitutes a block, which depends on the average amount of a given stock that is traded. Once a transaction has been decided upon, it must be channeled through an electronic system, which verifies that the trade is sufficiently large and that the price of the transaction is based on the volume-weighted average price. Trades that do not satisfy the size and price criteria are rejected by the system. The accepted trades are called applications and they must be declared to the market authorities immediately.

To gain insight when to trade applications, we gathered a continuous record of prices and quantities for 37 stocks traded between 1996 and 2003.² We first computed several statistics related to applications for each company.³ We report summary statistics of

²We use data purchased from the Paris Stock Exchange. For the analysis involving block trades, we used data covering 37 stocks between 1996 and 2003. For the optimal trading strategy, it is necessary to gather information about the order book which is a rather complex process. Hence, we consider liquidation strategies for five randomly chosen stocks but included in the main index. Data for this market was also used previously by Biais, Hillion, and Spatt (1995), who focus on the order flow dynamics.

³Unlike Keim and Madhavan (1995), we have no information if a trade was buy or sell initiated for

those averages in Table 1. Focusing on the column ‘Days with applic,’ we see that the company with the fewest block trades had a block trade on 76% of all days, whereas there was a block trade for each and every day for other companies. For a company selected at random, there are block trades 95% of the time on a given day, on average. A subdivision of the sample shows that the estimates are comparable over time. The next columns of the table show that applications represent only a small number of all trades (about 1.35%). However, the number of blocks traded on a given day can be as large as 30. The number of shares traded via blocks represents about 18% of all shares traded, on average. An average block trade is about 640,000 euros. The average block size can be as large as 2 million euros for some companies. Comparing these basic figures across the subsamples reveals that the proportion of block trades has decreased from 2000 to 2003. However, the size of the average block trade has increased over this same time period. The figures between 2000 and 2003 are typically twice as large as those observed between 1996 and 1999. All in all, the proportion of shares traded through applications is broadly constant through time.

The last two columns of Table 1 display the average and maximal amounts of euros traded for the 37 companies. As one would expect, the average amount of euros for an application is much larger than the amount traded in a regular trade. Comparing the average block value and the largest regular trades reveals an important feature: the amounts traded in blocks are smaller than the largest regular trades. This observation suggests that block trades are not the only way by which large orders can be executed. Many large trades appear to be channeled through the electronic system.

To further investigate the choice of the upstairs market versus the electronic market, we run a panel regression over 37 companies, with data covering the period 1996-2003. The dependent variable is the number of applications per day. The regression is:

$$\begin{aligned} nbApplic_{it} = \mu_i + a_0 nbTr_{t-1} + a_1 TV_{t-1} + a_2 vol_{t-1} \\ + a_3 R_{t-1} + a_4 R_{t-1}^- + a_5 R_{t-1}^{e-} + u_{it}, \end{aligned} \quad (1)$$

The past information used to describe the number of applications is the number of trading periods ($nbTr$), the total variance (TV), the volume traded (vol), the return (R), negative returns (R^-) and extremely negative returns (R^{e-}). A negative return is considered to be extreme (representing a crash) if it is below -4% . Table 2 presents the estimation results.

this database.

The table demonstrates that, on heuristic grounds, an investor may wish to perform a block trade subsequent to days with more trading periods and greater trading activity. However, the parameters on R^- and R^{e-} dummies indicate that, after negative market developments (which presumably represent bad news), the number of block trades decreases significantly. For the sake of expediency, a trader may sell through the electronic system in a sharply falling market, rather than seek a counterparty, which is potentially time consuming.

We also investigated the impact of a large positive return. Unlike a (negative) crash, we find that strong positive variation in the market is not followed by a decrease in liquidity or a decrease in the number of block trades.

To conclude this section, we note that block trades are a common and important way to liquidate large positions. However, investors also appear to channel large trades through the electronic market directly. Subsequent to negative market shocks, when block trades appear less frequent, investors may have a particular incentive to use an optimal trading strategy.

3 Optimal Trading: The Problem

The model's main building blocks are the trading strategy, the optimization problem and the price dynamics. The definition of trading strategy and the representation of the optimization problem are closely related to those adopted by Almgren and Chriss (2001) and Huberman and Stanzl (2000). We describe the mechanism of the optimal selling strategy, deferring the discussion of the optimal purchase strategy to a later section. Assume that an investor has a large position X in a security and that she wants to liquidate it completely by time T . A liquidation strategy is defined by a sequence of positive numbers $(x_{t_0}, \dots, x_{t_N})$, where $x_{t_i} \geq 0$ is the number of units the investor intends to hold at time t_i , for $i = 0, \dots, N$. The strategy is implemented at time t_0 , where $t_0 = 0 < t_1 < \dots < t_N = T$, and $x_{t_i} \geq x_{t_j}$ if $i < j$. Times t_i are assumed to be discretely and equally spaced.⁴ Given that initially the investor starts with X units of the security and she needs to liquidate the entire position by time T , it follows that $x_{t_0} = X$ and $x_{t_N} = 0$. The liquidation trajectory $(x_{t_0}, \dots, x_{t_N})$ can be expressed as a

⁴Since the liquidation strategies may find that a liquidation of 0 unit in some periods is optimal, and since we endogenize the number of liquidation periods, our assumption that the time between liquidation periods is constant does not constitute a limitation.

trade list $(n_{t_1}, \dots, n_{t_N})$, where n_{t_k} is the number of units sold immediately before time t_k . Hence, the variables x_{t_k} and n_{t_k} are related by

$$x_{t_k} = X - \sum_{j=1}^k n_{t_j}, \text{ for } k = 0, \dots, N.$$

This equation states that the investor's holding at time t_k is the initial holding, X , minus the sum of the quantities sold up to time t_k , $\sum_{j=1}^k n_{t_j}$. The liquidation strategy is based on the information available at time t_0 .⁵ The investor's optimization problem consists of finding the liquidation strategy $(x_{t_0}, \dots, x_{t_{N-1}})$, or the equivalent list $(n_{t_1}, \dots, n_{t_N})$, that maximizes the expected total income of liquidation, penalized by its variance, with the penalization factor denoted by η . We also assume affine transactions costs $K + n_t \kappa$. The parameters K and κ represent a fixed and a variable cost for each transaction, respectively. This leads us to the following optimization program:⁶

$$\begin{aligned} \max_{\{n_{t_1}, \dots, n_{t_N}\}} & E \left[\sum_{i=1}^N n_{t_i} (p_{t_i} - \kappa) \right] - \eta V \left[\sum_{i=1}^N n_{t_i} p_{t_i} \right] - NK, \\ \text{s.t.} & \sum_{i=1}^N n_{t_i} = X, \text{ and} \\ & 0 \leq n_{t_i} \leq X, \text{ for } i = 1, \dots, N. \end{aligned} \quad (2)$$

The optimization problem described in (2) assumes a specific number of trading periods N . To endogenize the number of trading periods, we solve the optimization problem for $N = 1, \dots, N^{\max}$, where $N^{\max}$ is large. We choose the optimal N to be the value that maximizes the objective function in (2). Note that the introduction of an affine transaction cost has two consequences. First, the variable cost $n_t \kappa$ corresponds to a change in the price. Second, the fixed cost NK only intervenes in the selection of the optimal number of trades N and not in the determination of n_t .

It is clear from (2) that the solution to the optimization problem depends on the dynamics of the trading price, p_{t_k} . We use this feature of the model to incorporate several microstructure features into the optimization, such as the time-varying impact of the direction and magnitude of the order flow. For instance, larger sales require larger discounts. Further, slow execution of the liquidation strategy implies more risk.

⁵At time t_0 , we define deterministic strategies, which determine the quantities to sell over the next N periods of time. Ideally, one would like to use adaptive strategies whereby the arrival of news affects the trading strategy. For the moment, no theoretical framework for such strategies appears to exist.

⁶The decision criterion is mean-variance. Experiments with other utility functions such as aversion towards downside risk reveal greater numerical complexity.

Regarding the purchase strategy, the problem is symmetric to the one discussed above. The objective function becomes:

$$\begin{aligned} \min_{\{n_{t_1}, \dots, n_{t_N}\}} & E \left[\sum_{i=1}^N n_{t_i} (p_{t_i} + \kappa) \right] + \eta V \left[\sum_{i=1}^N n_{t_i} p_{t_i} \right] + NK, \\ \text{s.t.} & \sum_{i=1}^N n_{t_i} = X, \text{ and} \\ & 0 \leq n_{t_i} \leq X, \text{ for } i = 1, \dots, N, \end{aligned} \quad (3)$$

where $x_{t_0} = 0$ and $x_{t_N} = X$. In this case, n_{t_i} denotes the number of units purchased immediately before time t_i .⁷ The interpretation and impact of K and κ are as before. Having understood the theoretical impact of transaction costs, we set $\kappa = K = 0$ from now on. It would be easy to modify the numerical implementation to account for these components.

We discuss the dynamics of the trading price in the following section, which incorporates separate dynamics for the buy and the sell sides of the order book.

3.1 The Price Dynamics

From now on, we drop the index k from the time t_k , denoting it by t . Therefore, time t represents the event time of the trade, and $t - 1$ represents the time of the trade occurring immediately before. In other words, we account for the time between trades, τ , in the estimation. The transaction price, p_t , is assumed to follow an extension of the dynamics proposed by Glosten and Harris (1988), Brennan and Subrahmanyam (1996), Madhavan, Richardson, and Roomans (1997), Huang and Stoll (1997), and Sadka (2006). The dynamics of the price movements are related to the dynamics of the fundamental value of the stock, denoted m_t . We define m_t as:

$$\begin{aligned} m_t = & \mu + m_{t-1} + \psi_t^+ (\mathbf{1}_{\{D_t=1\}} - E_{t-1} [\mathbf{1}_{\{D_t=1\}}]) - \psi_t^- (\mathbf{1}_{\{D_t=-1\}} - E_{t-1} [\mathbf{1}_{\{D_t=-1\}}]) \\ & + \lambda_t^+ (n_t - E_{t-1} [n_t]) \mathbf{1}_{\{D_t=1\}} - \lambda_t^- (n_t - E_{t-1} [n_t]) \mathbf{1}_{\{D_t=-1\}} + y_t, \end{aligned} \quad (4)$$

where $y_t \sim iid(0, \sigma_y^2)$.

⁷For institutional investors who seek to minimize tracking error with respect to some index, an alternative strategy may be to trade as many shares as possible right before the close of the market. Another interesting strategy would be to allow limit orders for the trading strategy. Such strategies would demand significant changes in the model. Given the complexity of currently available dynamic order books (see Goettler, Parlour, and Rajan, 2005), we leave this analysis to further research.

The variable m_t represents the expected value of the security, conditional on the information available at time t . For a given transaction at time t , the indicator variable D_t represents the direction of the order flow, which is defined as $+1$ if the trade is buy-initiated and -1 if it is sell-initiated. The variable n_t represents the order flow (i.e., the traded volume). The variable y_t is an exogenous shock representing the arrival of new information. Our model of price dynamics is innovative as it assumes differentiated responses for the buy and sell sides. Furthermore, we introduce a parameter μ , which describes market sentiment. This parameter represents a mean rate of drift and may be difficult to estimate; however, it may be useful in describing investors' expectations of bullish or bearish markets.

Equation (4) models the permanent effect of both private information, D_t and n_t , and public information, y_t , on price. The intuition behind this representation is that some traders have some private information about the price of the security, which is reflected in the decisions that they make. These decisions include the decision to buy or sell (i.e., the value of D_t), and the quantity to buy or sell (i.e., the value of n_t). Hasbrouck (1991a and b) and Foster and Viswanathan (1993), among others, have documented the presence of predictability in the order flow.⁸ To exclude the predictable part of the trade, equation (4) does not assume that the direction and magnitude of the order flow have an impact on the change in the fundamental value of the stock. Instead, only their innovation processes, represented respectively by, $\mathbf{1}_{\{D_t=1\}} - E_{t-1} [\mathbf{1}_{\{D_t=1\}}]$, $(\mathbf{1}_{\{D_t=-1\}} - E_{t-1} [-\mathbf{1}_{\{D_t=-1\}}])$, and $(n_t - E_{t-1} [n_t])$, respectively, have an impact. In (4), the parameters ψ_t and λ_t measure the permanent (fixed) and variable impacts on price. The superscripts $+$ and $-$ of ψ_t and λ_t indicate whether a trade is a buy or a sell order.

Following Glosten and Harris (1988), the dynamics of the transaction price depend on the transitory impact of the order flow and the direction of the order flow:

$$p_t = m_t + \bar{\psi}_t^+ \mathbf{1}_{\{D_t=1\}} - \bar{\psi}_t^- \mathbf{1}_{\{D_t=-1\}} + \bar{\lambda}_t^+ n_t \mathbf{1}_{\{D_t=1\}} - \bar{\lambda}_t^- n_t \mathbf{1}_{\{D_t=-1\}} + \xi_t, \quad (5)$$

where $\xi_t \sim iid(0, \sigma_\xi^2)$. This equation incorporates transient microstructure phenomena into the price process, generating a wedge between the fundamental value of the security, m_t , and the actually traded price. The parameter $\bar{\psi}_t$ can be interpreted as a fixed cost

⁸The predictability of prices is related to several empirical issues. For instance, the decisions of some investors to break large trades into small orders to reduce the price impact can create serial correlation in the order flow.

per order and the parameter $\bar{\lambda}_t$ as a variable (unitary) cost. Parameters $\bar{\psi}_t$ and $\bar{\lambda}_t$ are compatible with inventory cost models. Intuitively, these costs should affect the price only at the current time and in a transitory manner. The error ξ_t captures features that are not explicitly modeled, such as stochastic rounding induced by price discreteness. It is worth mentioning that in (4) and (5) the parameters, ψ_t , $\bar{\psi}_t$, λ_t , and $\bar{\lambda}_t$, are time-varying functions.⁹ A particular pattern of this time-varying impact is the well-illustrated feature in microstructure theory known as the U-shaped form of variables related to the order book.¹⁰

The estimation of the parameters is based on the set of trades (i.e. buy-initiated as well as sell-initiated trades), whereas we focus only on buy or sell strategies in the optimal trading strategy. Consequently, we assume that the predictable part of the series (captured by the terms $E_{t-1}[D_t]$ and $E_{t-1}[n_t]$) is negligible. In other words, the predictable component captures the effect of all other trades. Therefore, for a pure selling or buying strategy, the fundamental value of the stock evolves according to:

$$m_t = \begin{cases} \mu + m_{t-1} - \psi_t^- - \lambda_t^- n_t + \sigma_t \tau^{1/2} \omega_t, & \text{for a sale,} \\ \mu + m_{t-1} + \psi_t^+ + \lambda_t^+ n_t + \sigma_t \tau^{1/2} \omega_t, & \text{for a purchase,} \end{cases} \quad (6)$$

where $\omega_t \sim iid(0, 1)$, and the decision variable n_t denotes the quantity the investor decides to sell or buy at time $\tau t = \frac{T}{N}t$ for $t = 1, \dots, N$. It is convenient to define $\sigma_{y_t} = \sigma_t \tau$.

The trading price is given by:

$$p_t = \begin{cases} m_t - \bar{\psi}_t^- - \bar{\lambda}_t^- n_t + \xi_t, & \text{for a sale,} \\ m_t + \bar{\psi}_t^+ + \bar{\lambda}_t^+ n_t + \xi_t, & \text{for a purchase,} \end{cases} \quad (7)$$

which corresponds to (5) for a pure selling or buying strategy (i.e., $D_t = -1$ or $D_t = +1$).

We use the price dynamics defined in (6) and (7) to determine the optimal liquidation strategy, which corresponds to the solution to the optimization problem described by (2).

Before determining the optimal strategy, we must estimate the parameters of our model. We describe the estimation procedure in the next section. In Section 4.1, we

⁹This parametrization relaxes the assumption that the permanent and transitory impacts of the trade characteristics on the price are constant throughout the day, as assumed by Almgren and Chriss (2001), Dubil (2002), Hisata and Yamai (2000), and Sadka (2004).

¹⁰Empirical evidence of this fact is found in Hasbrouck (1991b) and Mönch (2004) and the references mentioned therein.

discuss the optimal trading strategy which results from the calibration.

3.2 Estimation of the Microstructure Model

Some filtering of the data is required before estimation: (i) the first trade of each day is eliminated, since the process generating the opening price (an auction) is different from the process generating the prices during the day and (ii) original volumes are seasonally adjusted. The rationale for this filter is that traded volume can be seen as the sum of a stochastic part and a deterministic part. The latter usually includes a time-of-the-day effect. This effect reflects systematic variation in traded volume during a normal day. Empirical evidence suggests that traded volume tends to be greatest just after opening and just before closing. We verified these features in our dataset. The seasonal adjustment procedure hopes to filter out this effect in order to keep only the stochastic effect. To correct for this effect, we adjust the raw traded volume series by the method proposed by Engle and Russell (1998).¹¹

3.2.1 Estimation - Step 1

The estimation is done in two steps. In the first step, we adjust and estimate parametric processes for the direction of the order book, D_t , and for the volume of the order flow, n_t . The estimated processes are used to infer the series $E_{t-1}[D_t]$ and $E_{t-1}[n_t]$. These series are used for the Generalized Method of Moments (GMM) estimation of (4) and (5).

In the first step, there is no consensus as to which model best describes the directions of the order book or the order flow. For instance, Hasbrouck (1991a and b) expresses quote-midpoints and order flows through a vector autoregressive process of order five. Based on this early model, Brennan and Subrahmanyam (1996) use five lags of price and order flow to estimate the unexpected order flow. Huang and Stoll (1997) model the direction of the order book as an autoregressive process of order one. Sadka (2006) estimates an $AR(5)$ process for the (signed) order flow.

¹¹We regress the logarithm of the raw volume (traded quantity) on time-of-the-day dummies. More specifically, the day is divided into K sub-periods and we consider the regression $\log(n_t) = \sum_{k=1}^K a_k x_{kt} + \varepsilon_t = a'x_t + \varepsilon_t$, where $x_{kt} = 1$ if the time of the trade t belongs to the intraday sub-period k for $k = 1, \dots, K$, and 0 otherwise. The seasonally adjusted series is defined by $\hat{n}_t = n_t \exp(-\hat{a}'x_t)$, where $\hat{a}$ denotes the OLS estimate of a .

We assume that the order flow, n_t , follows an autoregressive process and that the direction of the order flow, D_t , follows a Markov Chain process with state-space $S = \{-1, 0, 1\}$.

Therefore, the order flow is described by: $n_t = c + \varphi_1 n_{t-1} + \dots + \varphi_p n_{t-p} + \varepsilon_t$, with $\varepsilon_t \sim iid(0, \sigma^2)$. Once the parameters of the processes n_t (i.e., $c, \varphi_1, \dots, \varphi_p$) and D_t (i.e., the transition probabilities $\pi_{i,j} = Pr\{D_t = j \mid D_{t-1} = i\}$ for $i, j = -1, 0, 1$) are estimated, we calculate

$$\widehat{E}_{t-1}[n_t] = \widehat{c} + \widehat{\varphi}_1 n_{t-1} + \dots + \widehat{\varphi}_p n_{t-p}, \quad (8)$$

and

$$\widehat{E}_{t-1}[D_t | D_{t-1} = i] = -\pi_{i,-1} + \pi_{i,1}, \quad \text{for } i = -1, 0, 1. \quad (9)$$

Estimations of an AR model for n_t reveal different patterns, depending upon the stock that is used. For instance, using a Bayesian information criterion, we found that Orange's order flow is best described by an $AR(6)$ model.¹²

3.2.2 Estimation - Step 2

We use GMM to estimate the remaining set of parameters $(\psi_t^+, \psi_t^-, \bar{\psi}_t^+, \bar{\psi}_t^-, \lambda_t^+, \lambda_t^-, \bar{\lambda}_t^+, \bar{\lambda}_t^-, \sigma_{y_t}^2, \sigma_{\xi_t}^2)$ describing the dynamics of the expected and traded prices in (4) and (5). Following Madhavan, Richardson, and Roomans (1998), we take the first difference of p_t in (5), yielding

$$\begin{aligned} p_t - p_{t-1} = & m_t - m_{t-1} + \bar{\psi}_t^+ \mathbf{1}_{\{D_t=1\}} - \bar{\psi}_{t-1}^+ \mathbf{1}_{\{D_{t-1}=1\}} - \bar{\psi}_t^- \mathbf{1}_{\{D_t=-1\}} + \bar{\psi}_{t-1}^- \mathbf{1}_{\{D_{t-1}=-1\}} \\ & + \bar{\lambda}_t^+ n_t \mathbf{1}_{\{D_t=1\}} - \bar{\lambda}_{t-1}^+ n_{t-1} \mathbf{1}_{\{D_{t-1}=1\}} - \bar{\lambda}_t^- n_t \mathbf{1}_{\{D_t=-1\}} + \bar{\lambda}_{t-1}^- n_{t-1} \mathbf{1}_{\{D_{t-1}=-1\}} \\ & + \xi_t - \xi_{t-1}. \end{aligned}$$

¹²Autocorrelation was not detected in other series, such as Suez or Sodexho-Allianz. Others, such as Alcatel, required as many as 10 lags in the AR process. This autocorrelation pattern is not related to the size of the company nor is it related to liquidity, since all these stocks have large daily trading volume.

By plugging in $m_t - m_{t-1}$ from (4) and grouping some terms, we find that the dynamics of the actual trade price are given by

$$\begin{aligned}
p_t - p_{t-1} = & \mu + \psi_t^+ (\mathbf{1}_{\{D_t=1\}} - E_{t-1}[\mathbf{1}_{\{D_t=1\}}]) - \psi_t^- (\mathbf{1}_{\{D_t=-1\}} - E_{t-1}[\mathbf{1}_{\{D_t=-1\}}]) \\
& + \lambda_t^+ (n_t - E_{t-1}[n_t]) \mathbf{1}_{\{D_t=1\}} - \lambda_t^- (n_t - E_{t-1}[n_t]) \mathbf{1}_{\{D_t=-1\}} \\
& + \bar{\psi}_t^+ \mathbf{1}_{\{D_t=1\}} - \bar{\psi}_t^- \mathbf{1}_{\{D_t=-1\}} - \bar{\psi}_{t-1}^+ \mathbf{1}_{\{D_{t-1}=1\}} + \bar{\psi}_{t-1}^- \mathbf{1}_{\{D_{t-1}=-1\}} \\
& + \bar{\lambda}_t^+ n_t \mathbf{1}_{\{D_t=1\}} - \bar{\lambda}_t^- n_t \mathbf{1}_{\{D_t=-1\}} - \bar{\lambda}_{t-1}^+ n_{t-1} \mathbf{1}_{\{D_{t-1}=1\}} + \bar{\lambda}_{t-1}^- n_{t-1} \mathbf{1}_{\{D_{t-1}=-1\}} \\
& + y_t + \xi_t - \xi_{t-1}.
\end{aligned} \tag{10}$$

If the parameters $(\psi_t^+, \psi_t^-, \bar{\psi}_t^+, \bar{\psi}_t^-, \lambda_t^+, \lambda_t^-, \bar{\lambda}_t^+, \bar{\lambda}_t^-, \sigma_{y_t}^2, \sigma_{\xi_t}^2)$ are constant, we use (10) and define $u_t = y_t + \xi_t - \xi_{t-1}$ to find that

$$\begin{aligned}
u_t = & y_t + \xi_t - \xi_{t-1} \\
= & p_t - p_{t-1} - \mu - \left(\psi^+ + \bar{\psi}^+ \right) \mathbf{1}_{\{D_t=1\}} + \left(\psi^- + \bar{\psi}^- \right) \mathbf{1}_{\{D_t=-1\}} \\
& + \psi^+ E_{t-1} [\mathbf{1}_{\{D_t=1\}}] - \psi^- E_{t-1} [\mathbf{1}_{\{D_t=-1\}}] \\
& - \left(\lambda^+ + \bar{\lambda}^+ \right) n_t \mathbf{1}_{\{D_t=1\}} + \left(\lambda^- + \bar{\lambda}^- \right) n_t \mathbf{1}_{\{D_t=-1\}} \\
& + \lambda^+ E_{t-1} [n_t] \mathbf{1}_{\{D_t=1\}} - \lambda^- E_{t-1} [n_t] \mathbf{1}_{\{D_t=-1\}} \\
& + \bar{\psi}^+ \mathbf{1}_{\{D_{t-1}=1\}} - \bar{\psi}^- \mathbf{1}_{\{D_{t-1}=-1\}} + \bar{\lambda}^+ n_{t-1} \mathbf{1}_{\{D_{t-1}=1\}} - \bar{\lambda}^- n_{t-1} \mathbf{1}_{\{D_{t-1}=-1\}}.
\end{aligned} \tag{11}$$

From this equation, we notice that innovations u_t are linear functions of the parameters. We have $V[u_t] = V[y_t] + V[\xi_t] + V[\xi_{t-1}] = \sigma_y^2 + 2\sigma_\xi^2$, and $Cov[u_t, u_{t-1}] = Cov[y_t + \xi_t - \xi_{t-1}, y_{t-1} + \xi_{t-1} - \xi_{t-2}] = -Cov[\xi_{t-1}, \xi_{t-1}] = -\sigma_\xi^2$. Hence, the parameters σ_ξ^2 and σ_y^2 can also be identified from orthogonality conditions. To obtain the orthogonality conditions, let

$$\begin{aligned}
U_{t,1} &= [u_t \quad u_t \mathbf{1}_{\{D_t=1\}} \quad u_t \mathbf{1}_{\{D_t=-1\}} \quad u_t E_{t-1}[\mathbf{1}_{\{D_t=1\}}] \quad u_t E_{t-1}[\mathbf{1}_{\{D_t=-1\}}]]', \\
U_{t,2} &= [u_t n_t \mathbf{1}_{\{D_t=1\}} \quad u_t n_t \mathbf{1}_{\{D_t=-1\}} \quad u_t E_{t-1}[n_t] \mathbf{1}_{\{D_t=1\}} \quad u_t E_{t-1}[n_t] \mathbf{1}_{\{D_t=-1\}}]', \\
U_{t,3} &= [u_t \mathbf{1}_{\{D_{t-1}=1\}} \quad u_t \mathbf{1}_{\{D_{t-1}=-1\}} \quad u_t n_{t-1} \mathbf{1}_{\{D_{t-1}=1\}} \quad u_t n_{t-1} \mathbf{1}_{\{D_{t-1}=-1\}}]', \\
U_{t,4} &= [u_t^2 - (\sigma_y^2 + 2\sigma_\xi^2) \quad u_t u_{t-1} + \sigma_\xi^2]',
\end{aligned}$$

be column vectors. By stacking these conditions, we obtain the orthogonality conditions

$$E \begin{bmatrix} U_{t,1} \\ U_{t,2} \\ U_{t,3} \\ U_{t,4} \end{bmatrix} = 0, \tag{12}$$

which can be used for GMM estimation.¹³

3.3 The Optimal Sell Strategy

In this section, we solve the optimization problem described by (2) to derive the optimal liquidation strategy. These propositions follow Almgren and Chriss (2001).

Proposition 1 *When the price function is given by (7), the solution to the optimization problem described by (2) corresponds to the solution to the linear system*

$$c_k + \sum_{t=1}^{k-1} (-\lambda_t^- - b_t)n_t + (d_k - b_k)n_k - \sum_{t=k+1}^N b_k n_t = 0, \quad \text{for } t = 1, \dots, N,$$

where $c_t = -\left(\sum_{j=1}^t \psi_j^- + \bar{\psi}_t^- - t\mu\right)$, $b_t = 2\eta\tau \left(\sum_{j=1}^t \sigma_j^2\right)$, $d_t = -2\left(\lambda_t^- + \bar{\lambda}_t^- + \eta\sigma_{\xi_t}^2\right)$, and $(n_1, \dots, n_N)$ satisfies the constraints of the original problem.

Rather than expressing the solution in terms of n_t , it is possible to express it as a function of x_t . This function is defined in the following proposition.

Proposition 2 *When the price function is given by (7), the solution to the optimization problem described by (2) corresponds to the solution to the linear system*

$$a_t + b_t x_{t-1} + 2c_t x_t + b_{t+1} x_{t+1} = 0, \quad \text{for } t = 1, \dots, N-1,$$

where $a_t = (\mu + \bar{\psi}_t^- - \bar{\psi}_{t+1}^- - \psi_{t+1}^-)$, $b_t = \lambda_t^- + 2\bar{\lambda}_t^- + 2\eta\sigma_{\xi_t}^2$, $c_t = -\lambda_{t+1}^- - \bar{\lambda}_{t+1}^- - \bar{\lambda}_t^- - \eta\sigma_{y_{t+1}}^2 - \eta\sigma_{\xi_{t+1}}^2 - \eta\sigma_{\xi_t}^2$, $n_t = x_{t-1} - x_t$, and $(x_0, \dots, x_N)$ satisfies the constraints of the original problem.

The proof of Proposition 1 is relegated to the Appendix. The proof of Proposition 2 is analogous and not presented here. We use Proposition 1 to determine the optimal liquidation strategy. This strategy is easy to implement since, as the proof of Proposition 1 reveals, problem (2) is equivalent to solving the Phase-I of the simplex algorithm, as described for instance by Nocedal and Wright (1999).¹⁴

¹³An estimation takes less than 3 minutes in MATLAB even for samples containing significantly more than 50,000 observations.

¹⁴There are efficient algorithms that perform such computations. We used the implementation available in the MATLAB optimization toolbox. Determination of a trading strategy (optimal number of trades and actual trade sizes) typically takes less than a minute.

It is worth noting that (2) assumes a specific number of trades. To endogenize the optimal number of trading periods, we first obtain the optimal liquidation strategy, $(n_1^*, \dots, n_N^*)$, for $N = 1, \dots, N^{\max}$. Then, in a second step, we choose N^{opt} as the value of N that maximizes the objective function and retain $(n_1^*, \dots, n_{N^{\text{opt}}}^*)$ as the optimal liquidation strategy.

3.4 The Optimal Buy Strategy

Our result is similar to Proposition 1, but for a buy strategy.

Proposition 3 *When the price function is given by (7), the solution to the optimization problem described by (3) corresponds to the solution to the linear system*

$$c_k + \sum_{t=1}^{k-1} (\lambda_t^+ + b_t) n_t + (d_k + b_k) n_k + \sum_{t=k+1}^N b_t n_t = 0, \quad \text{for } t = 1, \dots, N,$$

where $c_t = \sum_{j=1}^t \psi_j^+ + \bar{\psi}_t^+ + t\mu$, $b_t = 2\eta\tau \left(\sum_{j=1}^t \sigma_j^2 \right)$, $d_t = 2 \left(\lambda_t^+ + \bar{\lambda}_t^+ \eta \sigma_{\xi_t}^2 \right)$, and $(n_1, \dots, n_N)$ satisfies the constraints of the original problem.

The following proposition gives the solution in terms of x_t .

Proposition 4 *When the price function is given by (7), the solution to the optimization problem described by (3) corresponds to the solution to the linear system*

$$a_t + b_t x_{t-1} + 2c_t x_t + b_{t+1} x_{t+1} = 0, \quad \text{for } t = 1, \dots, N-1,$$

where $a_t = (\mu + \psi_{t+1}^+ + \bar{\psi}_{t+1}^+ - \bar{\psi}_t^+)$, $b_t = -(\lambda_t^+ + 2\bar{\lambda}_t^+ + 2\eta\sigma_{\xi_t}^2)$, $c_t = \lambda_{t+1}^+ + \bar{\lambda}_{t+1}^+ + \bar{\lambda}_t^+ + \eta\sigma_{y_{t+1}}^2 + \eta\sigma_{\xi_{t+1}}^2 + \eta\sigma_{\xi_t}^2$, $n_t = x_{t-1} - x_t$, and $(x_0, \dots, x_N)$ satisfies the constraints of the original problem.

The proofs of Propositions 3 and 4 are similar to those of Propositions 1 and 2 and are not presented here.

4 Empirical Results

In this section, we use high frequency data to estimate the parameters ψ , $\bar{\psi}$, λ , $\bar{\lambda}$, σ_y , and σ_ξ , distinguishing the buy from the sell side.¹⁵ We use trades and quotes to infer

¹⁵The high frequency data is taken from the Paris Stock Exchange database. The original data contains information about all the buy and sell contracts, including date, price, and traded quantity,

the direction of the order flow, D_t , in addition to the following rule: If the trade price is larger than or equal to the last best ask price, then the trade is classified as buy-initiated (i.e., $D_t = +1$), if the trade price is smaller than or equal to the last best bid price, then the trade is classified as sell-initiated (i.e., $D_t = -1$), and if the trade is between the last best bid and the last best ask price, then the trade direction is assigned to 0.¹⁶

The estimation is performed using many stocks, although we present the results for only five randomly selected stocks: Alcatel, France Telecom, Orange, Sodexho Alliance, and Suez. The optimal liquidation strategy is established in three cases: (1) when the value of the parameters is constant; (2) when the parameter λ_t follows a hypothetical increasing function of time; and (3) when each parameter evolves continuously over time, the function governing the evolution being a spline approximation of the hourly estimates. In the first case, we also implement a sensitivity analysis to study how the optimal number of trading periods and the speed of liquidation are affected by changes in the value of the parameters.

4.1 Optimal Trading Strategy for Constant Parameters

We first assume that the parameters governing the price evolution in (4) and (5), ψ_t , $\bar{\psi}_t$, λ_t , $\bar{\lambda}_t$, $\sigma_{y_t}^2$, and $\sigma_{\xi_t}^2$, are constant throughout the day. For $\sigma_{y_t}^2$, this implies $\sigma_{y_t}^2 = \sigma^2\tau$, where τ is the time between two consecutive trades. We describe the parameter estimates we obtain from the two-step procedure described in Section 3.2 and then determine the optimal liquidation strategy following the procedure described in Section 3.3.

with a precision of 1 second. It also contains information about quotes, including date, best bid, best ask, depth at the best bid (maximum quantity to be bought at the best bid), and depth at the best ask (maximum quantity offered at the best ask).

¹⁶To evaluate the quality of this inference methodology, we compared the direction of the order flow obtained from the inference method with the actual direction of the order flow content in the order book. For all the stocks studied, the performance of the inference methodology was very good. The aforementioned methodology consistently classifies more than 96 percent of the trades as sell-initiated or buy-initiated, remaining undefined for less than 4 percent of the trades. It is worth clarifying that the classification methodology of Lee and Ready (1991) was developed for quote-driven markets and therefore does not apply to our case, since the Paris Stock Exchange is an order-driven market.

4.1.1 Estimation

In the first step of the estimation procedure, we adjust and estimate an autoregressive process for the order flow, n_t , and assume that the direction of the order flow, D_t , evolves according to a Markov Chain. In the second step, the other parameters (i.e., ψ , $\bar{\psi}$, λ , $\bar{\lambda}$, σ_y^2 , and σ_ξ^2) are estimated using GMM, where the set of orthogonality conditions is given by (12).¹⁷ We also performed tests of the restrictions where the buy and sell parameters were the same (that is, if $\vartheta^+ = \vartheta^-$ for $\vartheta \in \{\psi, \bar{\psi}, \lambda, \bar{\lambda}\}$). For the cases where we could not reject the null of equalities of the parameters, we present the same parameters for the buy and the sell sides.

Table 3 reports the parameter estimates for various companies. Intuitively, the value of the parameters ψ and λ should be positive.¹⁸ Indeed, a sell order (i.e., $D_t = -1$) typically reflects the belief that the price is high (the security is overvalued) and that it should decrease. Conversely, a buy order (i.e., $D_t = +1$) reflects the belief that the price is low (the security is undervalued) and should increase. The same interpretation applies to the parameter λ , which measures the impact of the direction of innovation of the order flow on prices.

In relation to the sign of the transient price impact parameters, the parameter $\bar{\psi}$ represents a fixed cost and is expected to be positive. The parameter $\bar{\lambda}$ represents a unitary cost. Intuitively, one might expect that this parameter is also positive. However, Sadka (2006) documents that this parameter may be negative, in which case there is a price discount.

Overall, we find that the parameters follow the expected sign, even though the variable transitory impact parameter, $\bar{\lambda}$, is insignificant for Sodexo and Suez. More specifically, we observe that the permanent impact of the innovation of the direction of the order book, ψ , and the transitory impact of the direction of the order book, $\bar{\psi}$, are positive and always significant. The parameter ψ can be interpreted as the expected change in the price after an unexpected change in the direction of the order flow (i.e., in $(D_t - E_{t-1}[D_t])$) and $\bar{\psi}$ is a fixed cost by order whose effect on price is only transitory. We observe that the signs of the estimates agree with the microstructure interpretation claiming that trades incorporate traders' private information. More specifically, traders decide to buy a security because they think it is undervalued and decide to sell because

¹⁷All prices were scaled as to yield comparable parameter estimates.

¹⁸The discussion holds for parameters with + or – superscripts.

they think it is overvalued.

The permanent effect of the direction of innovation of the order flow and the transitory effect of the direction of the order flow are captured respectively by λ and $\bar{\lambda}$, respectively. We observe that the former is positive and significant, meaning that a sell order has a negative impact on the price, while a buy order has a positive impact. The parameter $\bar{\lambda}$ is interpreted as the (transitory) marginal cost per unit. Its estimated value is negative, meaning that the seller of a certain amount of shares receives a discount for this sale.

4.1.2 Optimization

We determine the characteristics of the optimal trading strategy for the selected companies, assuming that the investor wishes to liquidate an order corresponding to 10% of the average daily traded volume. The available time horizon is $T = 8.5$ hours. We take $\eta = 4$ for the benchmark risk aversion. The parameters are those provided in Table 3.

A first illustration of the optimal trading strategy is given in Figure 1, where we focus on Orange. We observe that a large percentage of shares is traded immediately. The sell strategy is however more aggressive than the buy strategy (fewer trades are made and more shares are traded each time). Inspection of the parameters of Table 3 reveals that the magnitudes of the parameters of the sell side (superscripts $-$) are larger than those of the buy side (superscripts $+$). Hence, the price impact will be larger for the sell side, leading to a more cautious sale. In any case, it is interesting to note that the optimal trading strategy consists of a large initial trade, suggesting that the electronic system of this type is able to provide liquidity for large deals. One may wonder whether Orange is the only company for which such a trading strategy would hold. To answer this question, we present statistics for all companies under consideration in Table 4, focusing on an optimal liquidation. First, we observe similar patterns for all sales. The first trade involves between 72% and 83% of shares. Next, following Mönch (2004), we define duration as the average weighted time needed to trade a position

$$D = \sum_{i=1}^N (t_i - 9) \frac{n_i}{X}, \quad (13)$$

where t_i is the time that has passed since the trade was decided (measured in hours), n_i represents the size of a trade, and X is the initial position. Thus, duration is an indicator of the speed at which the trade is executed. It also represents the time at

which the number of shares that has been traded equals the number of shares that remain to be traded. As Table 4 documents, the duration of the various trades is, in all cases, under an hour, ranging from about 20 minutes to about 32 minutes. The total number of trading periods is between 5 and 7.

The next question that we address is which component of the microstructure model most strongly influences the expected revenue from the trading strategy. We consider the expected revenue relative to the revenue if one had sold all of the shares at one time and if one had no price impact at all. We have

$$\frac{1}{p_0 X} E \left[\sum_{t=1}^N n_t p_t \right] = 1 - \sum_{t=1}^N (\mu - \psi^-) \sum_{j=t}^N n_j - \sum_{t=1}^N \lambda^- n_t \sum_{j=t}^N n_j - \sum_{t=1}^N \bar{\psi}^- n_t - \sum_{t=1}^N \bar{\lambda}^- n_t^2.$$

The various components of revenue are presented in Table 4. We observe various effects. We note that the gains due to the direct price impact, be it transitory ($\bar{\psi}_t^-$) or permanent (ψ_t^-), are relatively small. We note that the unitary permanent impact (λ^-) is important and ranges from 1.3% to 20%. We also note that the transitory discount ($\bar{\lambda}^-$) ranges from about 0 for Suez to about 10.7% for Alcatel. This suggests that the optimal sale strategy involves a fraction of the trade, not because of an important direct impact, but because the optimizer tries to keep the cost of the permanent impact of a large sale under control. This effect is partially offset by the transitory impact of the sale.¹⁹ In the following subsection, we perform a comparative static exercise and study how the optimal selling strategy changes when the risk aversion of the price-impact parameters move away from the benchmark.

4.1.3 Sensitivity Analysis

We wish to discuss the effects of changing a parameter of the model. In Appendix 2, we consider the situation where parameters are kept constant and where an equal amount of shares are liquidated in each period.

Sensitivity to the risk aversion η . We change the value of the parameter η with respect to the benchmark from $\eta = 4$ to $\eta = 8$. The investor is expected to liquidate her position more quickly as she becomes more impatient, in order to avoid the risk

¹⁹We found very similar results for the buy strategy and for all measures reported in Table 4. The strong similarity is the reason we do not report the results. We found also very similar results for the strategy implemented in the case of continuously varying parameters.

of a potential price change. This fact is verified in Figure 2, where we represent the cumulative number of shares sold, by comparing the slopes of the curves denoted by GMM estimates and $\eta = 8$. This selling strategy permits the investor to avoid part of the exposure she faces due to price changes. Another consequence of the change in the value of the parameter η is the fact that the optimal number of trades increases from 5 to 6. This change allows for minimization of the price impact.

Sensitivity to the parameter ψ . We expect that the larger the value of the permanent impact parameter ψ , the smaller the optimal number of trading periods. The intuition is that each time the investor places an order, the price is expected to decrease permanently by ψ units. Thus, if the value of the parameter increases, the investor has more incentive to avoid its negative price impact by placing fewer orders. To evaluate this hypothesis, the parameter ψ is halved with respect to the benchmark. From Figure 2, we observe that the decrease in the value of the parameter ψ implies that the liquidation speed slightly increases at the beginning and decreases towards the end (see the slopes of the curves). As a consequence, the optimal number of trading periods increases from 5 to 6. Therefore, on average, the liquidation speed increases as larger lots are sold.

Figure 3 displays the optimal number of trading periods as a function of ψ . We observe that the optimal number of trading periods is decreasing in ψ and that its sensitivity to changes in the parameter value grows with smaller ψ . The value of the objective function is lower at the new parameterization. This change is a consequence of the increase in the fixed costs that is incurred each time an order is placed.

Sensitivity to the parameter $\bar{\psi}$. The parameter $\bar{\psi}$ represents a transitory fixed cost. When this parameter is constant over time, as is shown in Appendix 2, its effect on the objective function is represented by the term $-\sum_{t=1}^m \bar{\psi}_t n_t = -\bar{\psi}X$. Thus, $\bar{\psi}$ affects the objective function through a constant term, implying that changes in its value do not alter the optimal liquidation strategy. Rather, it only changes the level of the objective function.

Sensitivity to the parameter λ . An increase in the value of the parameter λ indicates that the permanent effect of each unit sold has a larger (negative) impact on price. If the unitary impact is larger, the investor might prefer to liquidate smaller quantities,

packaging the total initial position in more trades. This observation implies that the investor should increase the number of trading periods. We double the parameter λ in order to evaluate this hypothesis. As a consequence, the optimal number of trades rises from 5 to 7 and, in each of the first periods, the investor liquidates approximately 5% less shares than in the benchmark case.

Figure 2 displays the percentage that has been sold under the optimal liquidation strategy over time. At the beginning and end of the liquidation period, we observe that the investor liquidates more slowly than in the benchmark case, while in the middle she liquidates slightly faster. The duration decreases by minutes with respect to the benchmark, implying that the liquidation speed slightly increases given the parameterization.

Figure 3 displays the optimal number of trading periods as a function of λ . We observe that it is increasing in λ . We also observe that the number of trading periods is much more sensitive to the parameter λ than to the parameter ψ .

Sensitivity to the parameter $\bar{\lambda}$. An increase in the value of the parameter $\bar{\lambda}$ causes a reduction in its positive impact on price. This relationship exists because $\bar{\lambda}$ is negative and has a negative effect in (6). If the unitary (positive) impact on price is smaller, the investor might prefer to liquidate smaller quantities and to increase the number of trading periods. To evaluate this hypothesis, we increase the value of the parameter $\bar{\lambda}$ with respect to the benchmark from $\bar{\lambda} = -0.032 \cdot 10^{-3}$ to $\bar{\lambda} = -0.016 \cdot 10^{-3}$. As a consequence, the optimal number of trades increases from 5 to 6.

As discussed in the context of Table 4, the effect of $\bar{\lambda}$ on the total expected revenue is important. Changes in $\bar{\lambda}$ are therefore expected to have important consequences for the optimal selling strategy. Indeed, Figure 2 shows a much shorter duration of the overall sale as well as a larger number of sales. This finding corroborates the importance of the parameter $\bar{\lambda}$. From Figure 3, we find that the optimal number of trading periods is increasing in $\bar{\lambda}$.

Sensitivity to σ_y . We first note that σ_{y_t} is composed of τ the waiting time at time t and σ , the fundamental uncertainty of an asset's fundamental value. In discussing the impact of σ_y , we refer to the change of the average weighting time or equivalently, the impact of a change in σ .

As shown in Appendix 2, the contribution of σ_y to the objective function is of the order of magnitude $\sigma_y^2 N$. An increase in the parameter σ_y will decrease the objective

function because the price uncertainty of the shares that remain to sell increases. To compensate for this effect, the optimizer will seek an optimal number of trades that minimizes this negative impact by choosing a smaller N .

To further illustrate this point, we decrease the value of the parameter σ_y from $50.36 \cdot 10^{-3}$ (benchmark) to $25.18 \cdot 10^{-3}$. From Figure 2, we see that the liquidation speed decreases strongly during the first half of the day, where approximately 90 percent of the initial position is liquidated. The duration increases significantly. As a comparison of the benchmark curve and the new curve shows, the impact of a change in σ_y is quite dramatic.

Figure 3 displays the evolution of the optimal number of trading periods as a function of σ_y . We observe that they are strongly decreasing in σ_y . The consequence of a change in the variance is therefore significant. The optimal number of trading periods increases from 5 to 8.

Sensitivity to σ_ξ . Appendix 2 shows that the contribution to the objective function is of the order of magnitude $-\sigma_\xi^2/N$. An increase in the parameter σ_ξ^2 will decrease the objective function. Contrary to the previous discussion concerning σ_y , the optimizer will propose an increase in N in order to mitigate this negative impact.

In the numerical exercise, we increase the parameter σ_ξ from $\sigma_\xi = 4.046 \cdot 10^{-3}$ (benchmark) to $\sigma_\xi = 6.069 \cdot 10^{-3}$, keeping the other parameters at their benchmark levels. We observe that the number of trading periods increases from 5 to 6, and that the investor liquidates smaller quantities in the first periods.

From Figure 2, we conclude that the liquidation speed is about as sensitive to variation in the parameter σ_ξ as to variation in the parameter σ_y . Figure 3 displays the evolution of the optimal number of trading periods as a function of σ_ξ , revealing that the function is increasing in σ_ξ .

To summarize, N^{opt} is decreasing in ψ , invariant to $\bar{\psi}$, increasing in λ , increasing in $\bar{\lambda}$, decreasing in σ_y , and increasing in σ_ξ . Notably, changes in σ_y and σ_ξ have a rather important impact on the optimal trading strategy.

4.2 Optimal Trading Strategy for Time-Varying Parameters

To empirically explore the time-varying behavior of the parameters in our model, we estimate them for different hours of the day. Figure 4 displays the hourly estimated

values of the parameters for Orange a telecommunications company. We observe that, in general, the parameters are not constant throughout the day, suggesting that this fact needs to be captured by the model. There are significant differences in the parameters over a trading day. In this section, we discuss the consequences of a time-varying price impact function λ_t . We then will discuss the consequences of allowing the parameters to vary. The study of such a dynamic setting is particularly relevant in light of evidence suggesting that different dynamics apply at the close of the market. Our focus is on the parameter λ_t , which is particularly important because it captures the permanent price impact of the order flow. We determine the optimal liquidation when the time-varying parameter λ_t follows the increasing hypothetical pattern displayed in the top panel of Figure 5, while the other parameters remain at benchmark levels. This pattern corresponds to the specification $\lambda_t = a + bt + ct^2$, where $a = 0.257 \cdot 10^{-3}$, $b = -0.051 \cdot 10^{-5}$, $c = 10^{-6}$, and t is measured in hours with respect to 9 a.m. Intuitively, we expect that the investor liquidates larger quantities at the beginning of the day, when the price impact of the order flow is smaller, rather than late in the afternoon when it is larger.

The bottom panel in Figure 5 displays the corresponding optimal liquidation strategy. We observe that the optimization recommends a liquidation of the position in five trades as in the benchmark case. Importantly, the main effect of introducing an increasing function for the parameter λ_t is an increase in the liquidation speed. The total position is liquidated faster relative to the situation without time variation in the parameter. In particular, the position is liquidated during the early moments of trading, when the permanent variable impact on price of the order flow is small.

We conclude that taking even relatively shallow variation in the parameters into account may have important consequences for the market. We also performed an exercise where we allowed the parameters to change over time according to the dynamics in Figure 4. From the hourly estimates, we obtained continuous functions using a spline interpolation. We found that the position was liquidated in four trades and in less time.

4.3 Market Sentiment and Trading

An investor who plans to trade a significant position may have priors on the evolution of the market. For instance, she may wish to sell a portfolio in a falling market. In this section, we analyze the responses of our model to particular market conditions,

assuming that the investor uses prior information in the calibration of her model.²⁰ A first assumption could be the belief that markets will be subject to a relatively sharp fall. We parameterize this belief by assuming a decrease of $\mu = -5\%$ in the fundamental value during the day. As Case 2 in Table 5 indicates, this assumption will yield a larger first trade with respect to the benchmark in Case 1, but a greater liquidation duration. This result is due to the fact that the variance remains the same and, given the relatively small impact of a sale and the high value of volatility, even a relatively large price drop does not lead to a panic sale.

Next, we consider the scenario in which the investor accounts for her prior about the future evolution of the market and also uses realistic order flow parameters. Indeed, we estimated the parameters of the microstructure model $(\psi, \bar{\psi}, \lambda, \bar{\lambda})$ as well as the duration between trades τ corresponding to the five day with the largest price drop.²¹ In Table 5, under Case 3, we represent those figures corresponding to the situation where only μ and τ adjust to the new market conditions. Using the relative speed of the order flow, we find that the position should be liquidated much faster. For instance, in Case 2, we find that the initial duration of 0.53 hours for Alcatel is reduced to 0.25 hours.

Next, we assume that our investor increments her information set and also uses the $\psi, \bar{\psi}$ parameters corresponding to the worst possible day (Case 4). Unambiguously, the initial number of shares traded increases and the duration diminishes across all companies. For companies with large coefficients, the impact is as large as expected. Lastly, our investor uses the full information set: she believes that markets will fall sharply, that markets experience a strong increase in trading volume, resulting in a decrease in the time between trades τ , and that the parameters of the price dynamics switched to a bearish market sentiment. As Case 5 documents, the size of the first trade ranges between 73% and 90% of the amount to be traded. This order of magnitude was also found in Case 2. However, the duration is significantly shorter than for the benchmark Case 1 and the other situations. The only exception is France Telecom, for which the duration is slightly higher than that for Case 4. The reason for this difference is that the impact on the price due to volume is more important for some companies

²⁰In practice, trading floors are often oriented by an economist who will give indications if a market should be expected to rise or fall.

²¹These days are not necessarily consecutive. We also estimated the parameters for the five days with the largest price increases. The eventual results are qualitatively comparable with the results reported here.

and less for others.

As this section documents, market sentiment may be added to our model by incorporating either beliefs about the price behavior or by adjusting the calibration to a worst case scenario. Such market sentiment may significantly affect the optimal trading strategy.

5 Conclusion

In this paper, we consider an investor who desires to trade a large number of shares. Facing the choice between executing a block trade versus optimal liquidation through an electronic system, we show that an investor would choose block trading subsequent to high trading volumes, which presumably characterize liquid markets. Inversely, an investor would choose to trade through an electronic system after adverse market conditions (falling markets and subsequent to a crash).

We consider an optimal trading strategy through an electronic system. To do so, we extend an existing microstructure model to account for the possibility of differentiated parameters for the buy and sell sides. This microstructure model contains permanent and transitory components. In addition, we allow for quantity-related price impacts. After calibrating this model to different stocks, we run our optimal trading strategy.

Sell orders will be traded more aggressively than buy orders. A comparative static exercise demonstrates the importance of the various parameters and shows that changes in volatility have a particularly important impact. The observation that volatility plays an important role is corroborated in an exercise where we allow for market sentiment by assuming that the investor has bullish or bearish beliefs. Indeed, we show that the mere belief that markets will fall (which translates into a negative drift term, keeping all other parameters constant) will not significantly affect the trading strategy. Adjustments of the microstructure-model parameters and the volatility are also required to yield significant changes in the pattern of trades.

6 Appendices

6.1 Appendix 1: Proof of Proposition 1.

In this appendix, we obtain the solution to the optimization problem described in (2) when the price dynamics are given by (6) and (7). Replacing p_t from (7) in the total income of liquidation, $\sum_{t=1}^N n_t p_t$, we have

$$\sum_{t=1}^N n_t p_t = \sum_{t=1}^N n_t m_t + \sum_{t=1}^N n_t (-\bar{\psi}_t^- - \bar{\lambda}_t^- n_t + \xi_t). \quad (14)$$

Since (6) can be written as $m_t = m_{t-1} + h_t - \lambda_t^- n_t$, where $h_t = \mu - \psi_t^- + \sigma_t \tau^{1/2} \omega_t$, we obtain

$$\sum_{t=1}^N n_t m_t = m_0 X + \sum_{t=1}^N (h_t - \lambda_t^- n_t) \left(\sum_{j=t}^N n_j \right).$$

Replacing this result in (14), it follows that

$$\sum_{t=1}^N n_t p_t = m_0 X + \sum_{t=1}^N \left[(h_t - \lambda_t^- n_t) \left(\sum_{j=t}^N n_j \right) \right] - \sum_{i=1}^N \bar{\psi}_t^- n_t + \sum_{t=1}^N n_t (-\bar{\lambda}_t^- n_t + \xi_t).$$

Taking the expectation and calculating the variance of $\sum_{t=1}^N n_t p_t$, it holds that

$$\begin{aligned} E \left[\sum_{t=1}^N n_t p_t \right] &= p_0 X + \sum_{t=1}^N E[h_t] \left(\sum_{j=t}^N n_j \right) - \sum_{t=1}^N \lambda_t^- n_t \left(\sum_{j=t}^N n_j \right) \\ &\quad - \sum_{t=1}^N \bar{\psi}_t^- n_t - \sum_{t=1}^N \bar{\lambda}_t^- n_t^2, \end{aligned}$$

and

$$V \left[\sum_{t=1}^N n_t p_t \right] = \sum_{t=1}^N V[h_t] \left(\sum_{j=t}^N n_j \right)^2 + \sum_{t=1}^N n_t^2 \sigma_{\xi_t}^2.$$

Replacing $E \left[\sum_{t=1}^N n_t p_t \right]$ and $V \left[\sum_{t=1}^N n_t p_t \right]$ in the objective function (2), grouping common terms, and using the results: $E[h_t] = \mu - \psi_t^-$ and $V[h_t] = \sigma_t^2 \tau$, we have

$$\begin{aligned} E \left[\sum_{t=1}^N n_t p_t \right] - \eta V \left[\sum_{t=1}^N n_t p_t \right] &= p_0 X + \sum_{t=1}^N \left[(\mu - \psi_t^-) \left(\sum_{j=t}^N n_j \right) \right] \\ &\quad - \sum_{t=1}^N \lambda_t^- n_t \left(\sum_{j=t}^N n_j \right) - \sum_{t=1}^N \bar{\psi}_t^- n_t - \sum_{t=1}^N \bar{\lambda}_t^- n_t^2 \\ &\quad - \eta \sum_{t=1}^N \sigma_t^2 \tau \left(\sum_{j=t}^N n_j \right)^2 - \eta \sum_{t=1}^N \sigma_{\xi_t}^2 n_t^2. \end{aligned} \quad (15)$$

Taking the derivative with respect to n_k , we have

$$\begin{aligned}
\frac{\partial}{\partial n_k} \sum_{t=1}^N \left[(\mu - \psi_t^-) \left(\sum_{j=t}^N n_j \right) \right] &= \sum_{t=1}^k (\mu - \psi_t^-) = k\mu - \sum_{t=1}^k \psi_t^-, \\
\frac{\partial}{\partial n_k} \left[- \sum_{t=1}^N \lambda_t^- n_t \left(\sum_{j=t}^N n_j \right) \right] &= -2\lambda_k^- n_k - \sum_{t=1}^{k-1} \lambda_t^- n_t, \\
\frac{\partial}{\partial n_k} \left[- \sum_{t=1}^N \bar{\psi}_t^- n_t \right] &= -\bar{\psi}_k^-, \\
\frac{\partial}{\partial n_k} \left[- \sum_{t=1}^N \bar{\lambda}_t^- n_t^2 \right] &= -2\bar{\lambda}_k^- n_k, \\
\frac{\partial}{\partial n_k} \left[-\eta \sum_{t=1}^N \sigma_t^2 \tau \left(\sum_{j=t}^N n_j \right)^2 \right] &= -\sum_{t=1}^N \left(\sum_{j=1}^{\min(t,k)} 2\eta \tau \sigma_j^2 \right) n_t, \\
\frac{\partial}{\partial n_k} \left[-\eta \sum_{t=1}^N \sigma_{\xi_t}^2 n_t^2 \right] &= -2\eta \sigma_{\xi_k}^2 n_k.
\end{aligned}$$

Thus, the total derivative of $L = E \left[\sum_{t=1}^N n_t p_t \right] - \eta V \left[\sum_{t=1}^N n_t p_t \right]$ is given by

$$\frac{\partial L}{\partial n_k} = - \sum_{t=1}^k \psi_t^- + k\mu - 2\lambda_k^- n_k - \sum_{t=1}^{k-1} \lambda_t^- n_t - \bar{\psi}_k^- - 2\bar{\lambda}_k^- n_k - \sum_{t=1}^N \left(\sum_{j=1}^{\min(t,k)} 2\eta \tau \sigma_j^2 \right) n_t - 2\eta \sigma_{\xi_k}^2 n_k,$$

for $k = 1, \dots, N$. Therefore, the solution to the optimization problem described in (2) is given by the solution to this system of linear equations, subject to the restrictions: $\sum_{t=1}^N n_t = X$, and $0 \leq n_k \leq X$ for $k = 1, \dots, N$. It follows that the optimization problem (2) for the price dynamics boils down to a solution of the Phase-I of the Simplex algorithm. In other words, we are seeking the integers n_k satisfying

$$A\mathbf{n} = \mathbf{a}, \text{ and}$$

$$0 \leq n_i \leq X, \text{ for } i = 1, \dots, N,$$

where A is an $(N+1, N)$ matrix, $\mathbf{a}$ is a vector of dimension $N+1$, and the decision vector $\mathbf{n}$ is of dimension N . Specifically, A , $\mathbf{a}$, and $\mathbf{n}$ are defined as

$$A = \begin{bmatrix} d_1 - b_1 & -b_1 & -b_1 & \cdots & -b_1 & -b_1 \\ -\lambda_1 - b_1 & d_1 - b_1 & -b_2 & \cdots & -b_2 & -b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ -\lambda_1 - b_1 & -\lambda_2 - b_2 & -\lambda_3 - b_3 & \cdots & -\lambda_{N-1} - b_{N-1} & -b_N \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{bmatrix},$$

$\mathbf{a} = (c_1, \dots, c_N, X)'$, and $\mathbf{n} = (n_1, \dots, n_N)'$. Very efficient numerical algorithms can be applied to solve this problem.

6.2 Appendix 2: Comparative Statics.

In order to better understand the patterns of the optimal number of trades as the parameters change, we consider a simplified selling strategy with equally weighted trades and constant parameters. For notational ease, we have dropped the superscript “—” from the parameters. Decomposing $E \left[\sum_{t=1}^N n_t p_t \right] - \eta V \left[\sum_{t=1}^N n_t p_t \right]$ from equation (15) and replacing n_t by X/N , we obtain the influence of each parameter:

$$\begin{aligned}
\sum_{t=1}^N \left[(\mu - \psi_t) \left(\sum_{j=t}^N n_j \right) \right] &= \sum_{t=1}^N \left[(\mu - \psi) \left(\sum_{j=t}^N \frac{X}{N} \right) \right] \\
&= (\mu - \psi) \frac{X}{N} \sum_{t=1}^N (N - t + 1) \\
&= \frac{1}{2} X (\mu - \psi) (N + 1), \\
-\sum_{t=1}^N \lambda_t n_t \left(\sum_{j=t}^N n_j \right) &= -\sum_{t=1}^N \lambda \frac{X}{N} \left(\sum_{j=t}^N \frac{X}{N} \right) = -\lambda \left(\frac{X}{N} \right)^2 \sum_{t=1}^N (N - t + 1) \\
&= -\frac{1}{2} \lambda X^2 \frac{N + 1}{N}, \\
-\sum_{t=1}^N \bar{\psi}_t n_t &= -\sum_{t=1}^N \bar{\psi} \frac{X}{N} = -X \bar{\psi}, \\
-\sum_{t=1}^N \bar{\lambda}_t n_t^2 &= -\sum_{t=1}^N \bar{\lambda} \left(\frac{X}{N} \right)^2 = -X^2 \bar{\lambda} \frac{1}{N}, \\
-\eta \sum_{t=1}^N \sigma_t^2 \tau \left(\sum_{j=t}^N n_j \right)^2 &= -\eta \sum_{t=1}^N \sigma^2 \tau \left(\sum_{j=t}^N \frac{X}{N} \right)^2 = -\eta \sum_{t=1}^N \sigma^2 \tau \left[\frac{X}{N} (N - t + 1) \right]^2 \\
&= -\eta \sigma^2 \tau \left(\frac{X}{N} \right)^2 \sum_{t=1}^N [N^2 - 2N(t - 1) + (t - 1)^2] \\
&= -\frac{1}{6} \eta \sigma^2 \tau X^2 \frac{(N + 1)(2N + 1)}{N}, \\
-\eta \sum_{t=1}^N \sigma_{\xi_t}^2 n_t^2 &= -\eta \sum_{t=1}^N \sigma_{\xi}^2 \left(\frac{X}{N} \right)^2 = -\eta \sigma_{\xi}^2 \frac{X^2}{N}.
\end{aligned}$$

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Captions

Table 1: This table presents descriptive statistics of block trades (called ‘applications’) for 37 companies. Mean, min, max, std represent the average, minimum, maximum, and standard deviation, respectively, over the average of a given statistic computed for 37 companies. The first block of figures refers to the entire sample (1996-2003) and the lower two blocks to subsamples.

Table 2: This table reports the estimates from a panel regression relating the number of applications on a given day to several explanatory variables. The variables are as follows: *nbApplic*, the daily number of block trades; *nbTr*, the total number of trading periods; *V*, a measure of daily realized volatility; *Vol*, the daily traded volume (in Euros); *R*, the daily return for a given company; R^- , negative returns, and R^{e-} denoting extremely negative returns (worse than -4%). The last two explanatory variables are included to account for leverage effects and extreme market corrections.

Table 3: This table reports GMM estimates of the model, keeping parameters in (4) and (5) constant. The parameters have been multiplied by 10^3 . The numbers in parentheses represent t-statistics for the null hypothesis that a parameter is equal to 0. The *J* statistic is a broad measure of goodness of fit. Parameters labeled with “+” correspond to the sell side and those labeled with “−” to the buy side. ψ and λ are permanent fixed costs respectively variable costs. $\bar{\psi}$ and $\bar{\lambda}$ represent transitory fixed costs and variable costs, respectively. σ_ξ is the standard deviation of the rounding error. σ_y is the standard deviation of the permanent component of the prices. The total standard deviation is given by $\tau^{1/2}\sigma_y$, where τ is the time that passes between two given trades. The model is estimated by considering the continuous flow of transactions. The samples contain at least 50,000 observations.

Table 4: This table reports characteristics of the optimal liquidation strategy with constant parameters. n_{t_1} is the number of shares sold at t_1 as a percentage of the total number of shares to be liquidated. Duration is defined in equation (13). N^{opt} is the optimal number of trading periods in the sales strategy. We have the following definitions:

$$\begin{aligned} \psi^- \text{ effect} &= 100 \left[\sum_{t=1}^N \psi_t^- \left(\sum_{j=t}^N n_j \right) \right] / p_0 X, \quad \bar{\psi}^- \text{ effect} = 100 \left[\sum_{t=1}^N \bar{\psi}_t^- n_t \right] / p_0 X, \\ \lambda^- \text{ effect} &= 100 \left[\sum_{t=1}^N \lambda_t^- n_t \left(\sum_{j=t}^N n_j \right) \right] / p_0 X, \quad \bar{\lambda}^- \text{ effect} = 100 \left[\sum_{t=1}^N \bar{\lambda}_t^- n_t^2 \right] / p_0 X. \end{aligned}$$

Table 5: This table reports characteristics of the optimal liquidation strategy under alternative scenarios. n_{t_1} , duration, and N^{Opt} are defined as in Table 4. Case 1 is the benchmark case that was considered in Table 4. In Case 2, the investor incorporates beliefs about the evolution of the price over the next day given a market crash of -5% . In Case 3, the investor accounts for an increased number of trading periods during agitated days. In Case 4, the investor believes that the impacts from ψ and $\bar{\psi}$ correspond to bad days, as well as to τ . In Case 5, the investor believes a crash will take place. All parameters are adjusted to reflect worst case scenarios and τ is set to a short value, corresponding to a day with high trading volume.

Figure 1: We present the optimal amounts of shares to be traded over the day for Orange. The investor must trade 6966 shares of stock (10% of the average daily volume). Stars represent the number of shares sold and triangles represent the number of shares bought.

Figure 2: We trace the cumulative amount of shares sold for various parameter settings. The company considered is Orange. The benchmark case is defined by parameters $X = 6966$, $T = 8.5$ hours, $\eta = 4$, $\psi^- = 2.199 \cdot 10^{-3}$, $\bar{\psi}^- = 5.448 \cdot 10^{-3}$, $\lambda^- = 0.042 \cdot 10^{-3}$, $\bar{\lambda}^- = -0.032 \cdot 10^{-3}$, $\sigma_y = 50.357 \cdot 10^{-3}$, and $\sigma_\xi = 4.046 \cdot 10^{-3}$. The benchmark curve is labeled GMM. Deviations are obtained by changing one parameter after another. η increased to 8, ψ , $\bar{\lambda}$, σ_y are halved, λ is doubled and σ_ξ increased by a factor of 1.5. We use the average time that passed between trades for τ .

Figure 3: The various panels represent the effect of changing one of the microstructure parameters on the number of daily sell orders. The company considered is Orange. We use the numbers presented in Figure 2 as benchmark parameters. All parameters have been multiplied by 10^3 .

Figure 4: The evolution of the various parameters over the day is presented here. The stars represent the point estimates and the dashed lines correspond to the 90% confidence interval.

Figure 5: The upper plot presents the function $\lambda(t)$ as estimated over the day and the lower plot presents the optimal liquidation over a given day.

	days with applic. (%)	trades that are applic (%)	nb of applic per day	shares traded in applic (%)	avg volume traded per applic	avg volume traded per non applic	max volume traded per non applic
Total sample (1996-2003)							
mean	94.73	1.35	10.25	17.93	638678.6	17965.6	845628.1
min	76.01	0.94	3.81	6.22	32903.8	2835.4	268876.2
max	100.00	1.97	30.19	21.14	1937910.0	59416.6	2940872.6
std	6.45	0.21	5.99	2.72	324377.4	9511.2	605623.1
1st sub-period (1996-1999)							
mean	93.97	1.86	8.95	19.41	282080.2	12614.2	543762.1
min	61.11	1.18	1.88	8.52	26664.2	2793.5	108706.6
max	100.00	3.05	21.43	24.04	1139937.1	44718.1	2279112.8
std	8.25	0.33	4.57	3.12	209676.1	8763.9	507477.2
2nd sub-period (2000-2003)							
mean	95.18	0.87	11.46	16.50	921804.2	21692.6	649778.4
min	54.91	0.55	2.73	3.95	53773.5	2975.5	150493.3
max	100.00	1.44	38.77	22.61	2448277.8	68817.5	2940872.6
std	8.54	0.19	8.28	2.85	482128.8	11725.9	535714.8

Table 1: Descriptive statistics on block trades

Variable	Parameter	p-value	Parameter	p-value
$nbTr_{t-1}$	0.0020	0.00	0.0034	0.00
$nbTr_{t-2}$	0.0011	0.00	-	-
V_{t-1}	0.0009	0.74	0.0051	0.04
V_{t-2}	0.0080	0.00	-	-
vol_{t-1}	0.0159	0.00	0.0226	0.00
vol_{t-2}	0.0159	0.00	-	-
R_{t-1}	0.2656	0.00	0.2188	0.00
R_{t-2}	0.0053	0.89	-	-
R_{t-1}^-	-0.5942	0.00	-0.5315	0.00
R_{t-2}^-	-0.2176	0.00	-	-
R_{t-1}^{e-}	-0.2369	0.00	-0.2155	0.00
R_{t-2}^{e-}	-0.0756	0.19	-	-
$\bar{R}^2$	0.259		0.255	
(Nobs, Nvars)	(73257, 49)		(73294, 43)	

Table 2: Parameter estimates of the model for the number of applications

	Alcatel	France Tel	Orange	Sodexho	Suez
ψ^+	1.246 (11.00)	4.885 (15.09)	2.243 (5.77)	11.033 (33.39)	5.107 (76.55)
ψ^-	1.461 (12.72)	4.690 (14.84)	2.199 (5.71)	11.033 (33.39)	5.107 (76.55)
$\bar{\psi}^+$	4.151 (54.10)	7.201 (31.07)	4.067 (14.66)	8.992 (25.20)	5.432 (88.23)
$\bar{\psi}^-$	4.373 (55.86)	8.154 (36.13)	5.448 (19.72)	8.992 (25.20)	5.432 (88.23)
λ^+	0.054 (7.53)	0.122 (8.40)	0.052 (5.87)	0.180 (2.06)	0.120 (2.45)
λ^-	0.048 (5.38)	0.042 (2.93)	0.042 (3.95)	0.180 (2.06)	0.120 (2.45)
$\bar{\lambda}^+$	-0.035 (-5.49)	-0.049 (-4.85)	-0.026 (-3.56)	-0.046 (-0.58)	-0.002 (-0.31)
$\bar{\lambda}^-$	-0.031 (-3.91)	-0.026 (-2.11)	-0.032 (-3.14)	-0.046 (-0.589)	-0.002 (-0.31)
σ_y	22.446 (13.71)	62.655 (15.48)	50.357 (14.57)	382.718 (13.30)	76.232 (14.47)
σ_ξ	2.543 (93.17)	8.347 (88.26)	4.046 (64.15)	16.990 (24.57)	7.461 (67.19)
J stat	5.832	1.732	1.088	2.107	1.200
p value	0.323	0.885	0.955	0.550	0.753

Table 3: GMM estimates of the model with constant parameters

	Alcatel	France Tel	Orange	Sodexho	Suez
n_{t_1}	83.2412	72.9076	76.8118	78.4821	72.5812
Duration D	0.3398	0.4497	0.5072	0.4618	0.5316
N^{opt}	5	7	5	5	6
ψ^- effect	0.0397	0.0395	0.0453	0.0644	0.0425
$\bar{\psi}^-$ effect	0.1020	0.0449	0.0726	0.0412	0.0328
λ^- effect	19.5409	4.2284	3.8517	1.3365	4.4518
$\bar{\lambda}^-$ effect	-10.7213	-1.6088	-1.9920	-0.2709	-0.0523

Table 4: Characteristics of the optimal liquidation strategy with constant parameters

		Alcatel	France Tel	Orange	Sodexho	Suez
Case 1	n_{t_1}	83.241	72.908	76.812	78.482	72.581
	Duration D	0.340	0.450	0.507	0.462	0.532
	N^{opt}	5	7	5	5	6
Case 2	n_{t_1}	87.424	73.612	90.677	83.837	74.704
	Duration D	0.534	0.722	0.396	0.687	0.857
	N^{opt}	2	4	2	2	3
Case 3	n_{t_1}	68.716	47.896	60.171	55.741	39.945
	Duration D	0.246	0.240	0.286	0.225	0.184
	N^{opt}	4	8	3	5	3
Case 4	n_{t_1}	76.576	71.768	81.638	69.113	59.252
	Duration D	0.229	0.142	0.106	0.165	0.104
	N^{opt}	3	5	3	4	7
Case 5	n_{t_1}	87.404	73.608	90.688	83.835	74.709
	Duration D	0.158	0.157	0.078	0.135	0.109
	N^{opt}	2	4	2	2	3

Table 5: Characteristics of the optimal liquidation strategy under alternative scenarios

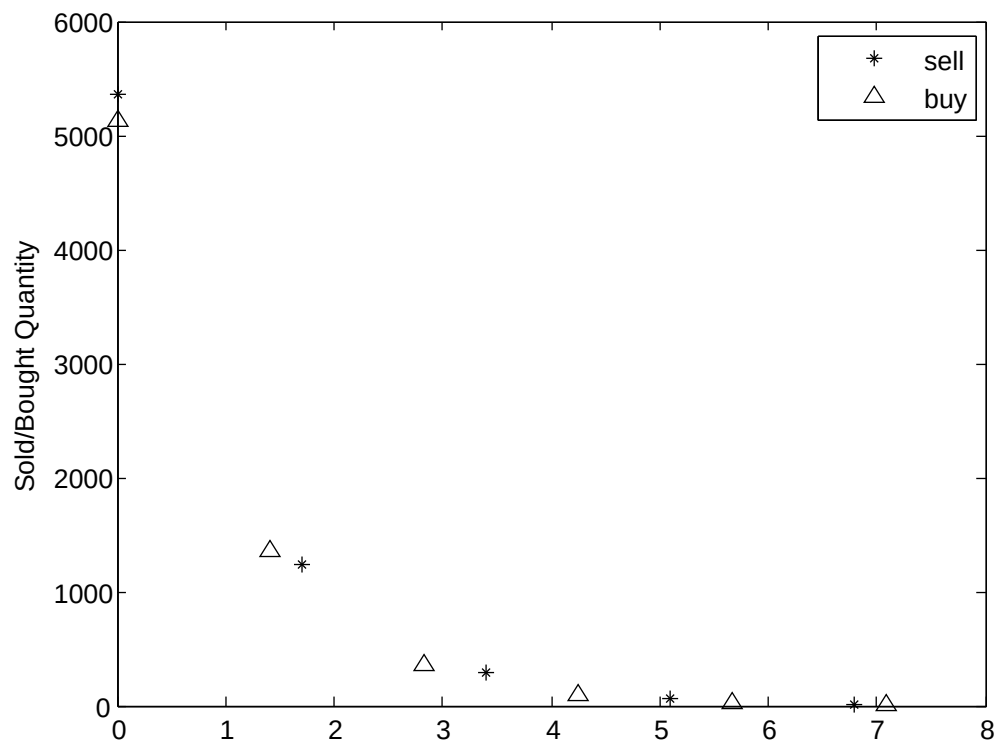


Figure 1: Optimal liquidation/purchase strategy

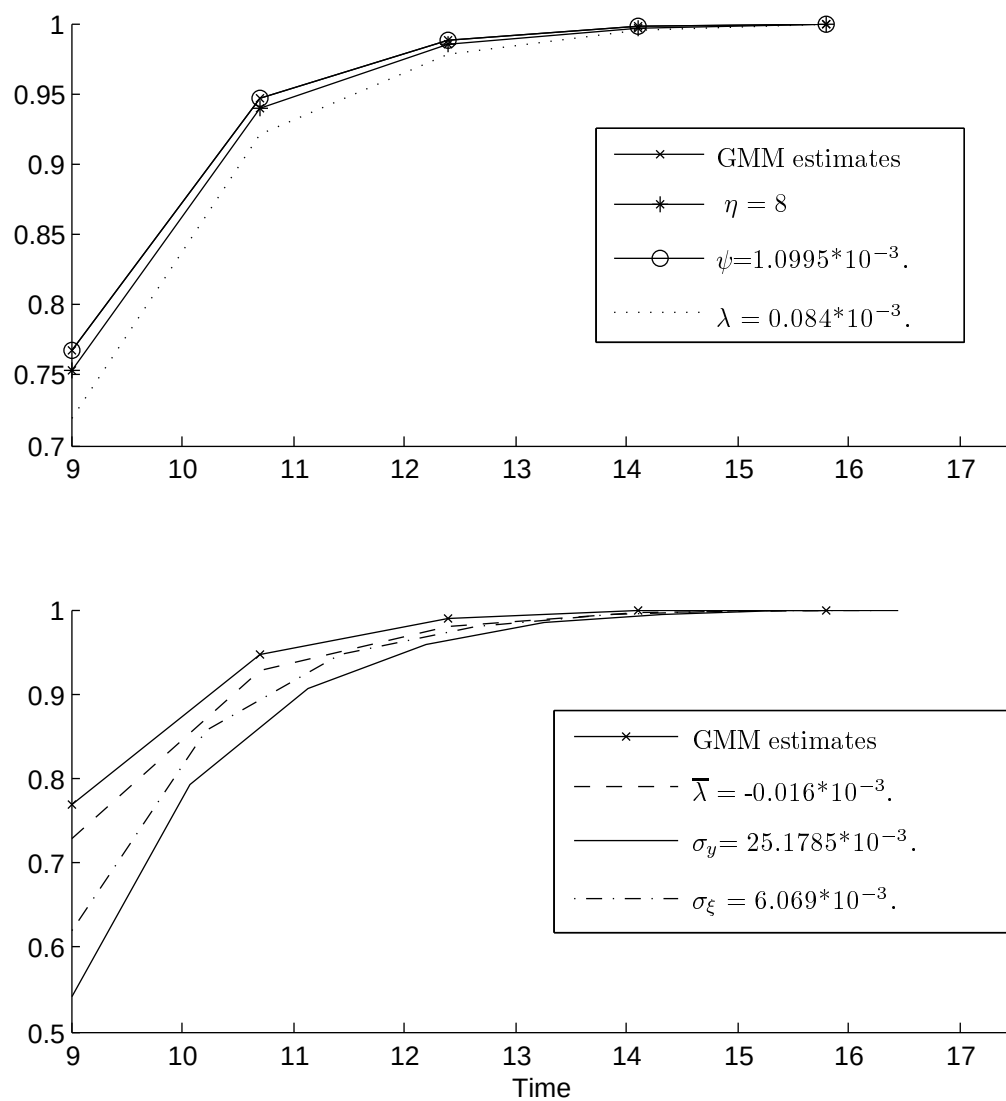


Figure 2: Sold percentage of the initial position

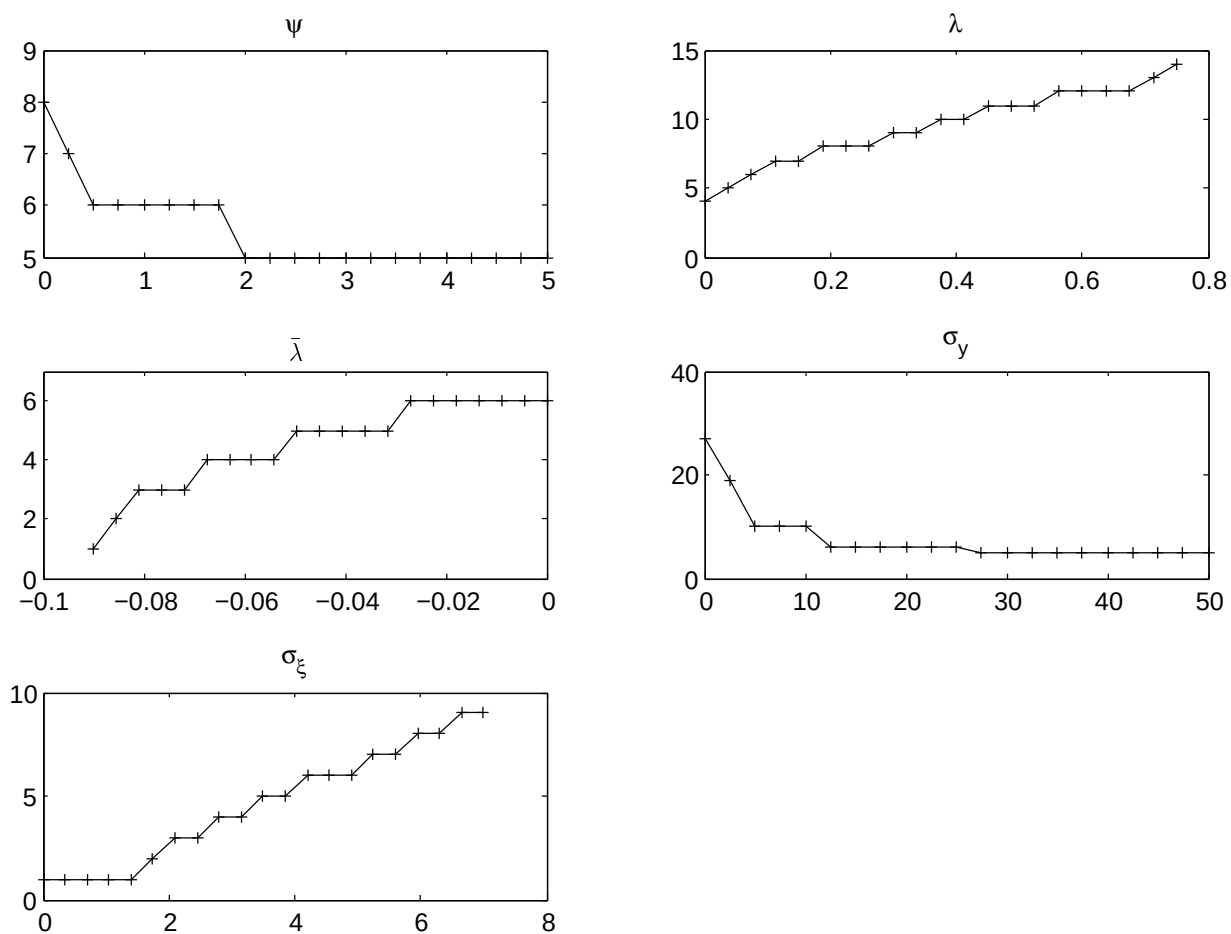


Figure 3: Number of daily trades depending on microstructure parameters

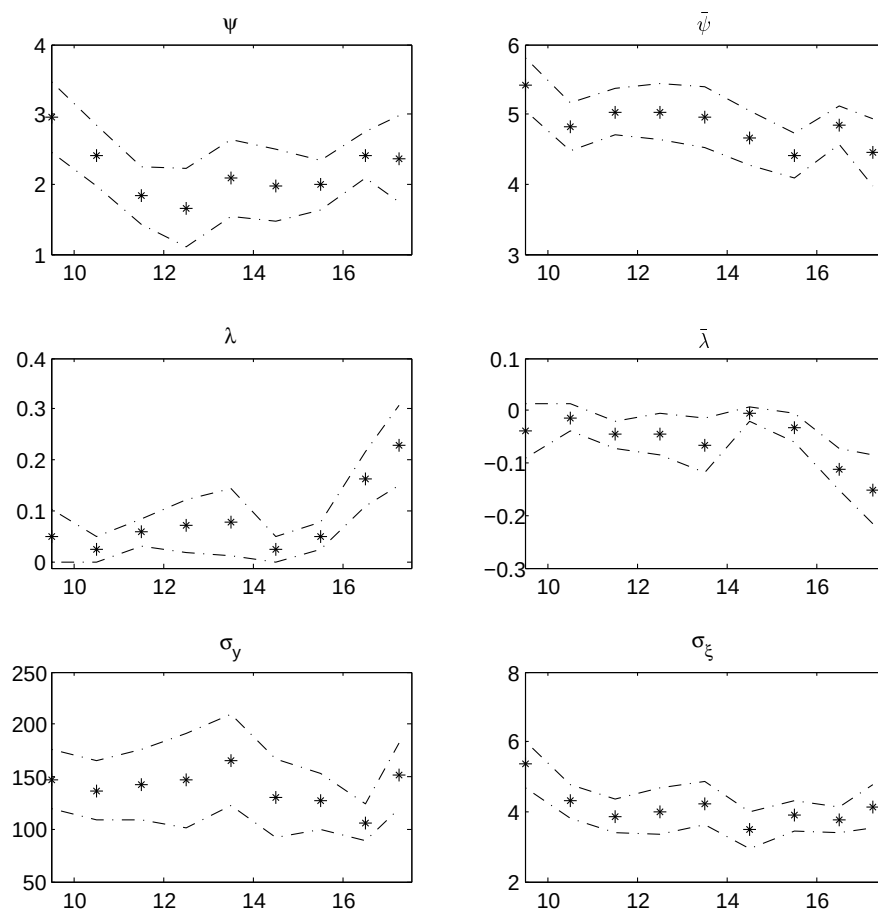


Figure 4: Evolution of the parameters over the day

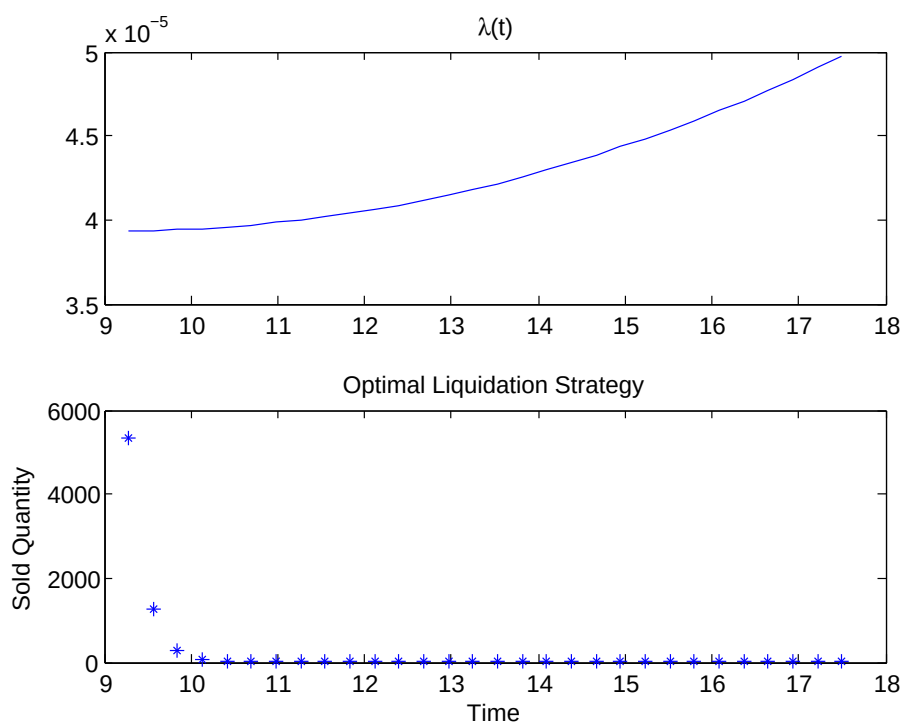


Figure 5: Function λ_t over the day and optimal liquidation strategy

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